(iv) $\dfrac{(\alpha+\beta)^2-2\alpha\beta+3(\alpha+\beta)}{\alpha\beta}$ 2.(i) $\frac{7}{5}$ (ii) $\frac{29}{10}$ (iii) $\frac{13}{6}$

3.(i) $x^2-44x+16=0$ (ii) $x^2-3x-1=0$ (iii) $x^2-24x-64=0$

4. $-15, 15$ 5. $-24, 24$ 6. -36

Exercise 3.15

1. ₹6500, ₹1250 2. y=8, x=4 3. y=4.5, x=15, 4. y=18 minutes, 10 pipes
5. 360, ₹30 6. ₹90, 10 hrs

Exercise 3.16

1.(i) Real and unequal roots (ii) Real and equal roots (iii) No real roots
(iv) Real and unequal roots (v) Real and equal roots (vi) Real and unequal roots
2. –3, 4 3. No real roots 4. -1
5. $-4, 1$ 6. $-2, 7$ 7. $-1, 3$ 8. $-2, 3$

Exercise 3.17

1.(i) 16 (ii) 4×4 (iii) $\sqrt{7}, \frac{\sqrt{3}}{2}, 5, 0, -11, 1$

2. $1\times18, 2\times9, 3\times6, 6\times3, 9\times2, 18\times1$ and $1\times6, 2\times3, 3\times2, 6\times1$

3.(i) $\begin{pmatrix}1&3&5\\0&2&4\\1&1&3\end{pmatrix}$ (ii) $\begin{pmatrix}\frac{8}{3}&9&\frac{64}{3}\\9&\frac{64}{3}&\frac{125}{3}\\\frac{64}{3}&\frac{125}{3}&72\end{pmatrix}$ 4. $\begin{pmatrix}5&1&3\\4&-7&8\\3&9&2\end{pmatrix}$

5. $\begin{pmatrix}-\sqrt{7}&\sqrt{5}&-\sqrt{3}\\3&-2&5\end{pmatrix}$ 7.(i) 3,12,3 (ii) 4,2,0 or 2,4,0 (iii) 2,4,3

Exercise 3.18

3. $\begin{pmatrix}5&0\\\frac{3}{2}&\frac{9}{2}\end{pmatrix}, \begin{pmatrix}2&0\\\frac{3}{2}&\frac{1}{2}\end{pmatrix}$ 4.(i) $\begin{pmatrix}7&-17&-37\\-39&-11&-26\end{pmatrix}$ (ii) $\begin{pmatrix}-63&-15&-45\\15&-27&-60\end{pmatrix}$

5.(i) $4, -10, 12$ (ii) $-10, 14, 10$ 6. 4,6 7. 4 8. $-1, 5$ and $-2, 4$

Exercise 3.19

1. $3\times3, 4\times2, 4\times2, 4\times1, 1\times3$ 2. $p\times r$, not defined 3. 7,10

4. $\begin{pmatrix}12&19\\10&3\end{pmatrix}, \begin{pmatrix}-10&-4\\24&25\end{pmatrix}$, $AB\neq BA$

Exercise 3.20

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
(D)	(A)	(B)	(A)	(B)	(C)	(D)	(B)	(C)	(C)	(B)	(A)	(B)	(D)	(B)	(B)	(D)	(B)	(C)	(A)

Unit exercise-3

1. 6,2,1 2. 42,78,30 3. 153 4. $(ky+x)(k^2x^2-y^2)$ 5. x^2+2x+1 6. (i) x^a-2
(ii) $-x+\frac{5}{2}$ 7. $\dfrac{(p+q+r)^2}{2qr}$ 8. 11 hrs, 22 hrs, 33 hrs 9. $|17x^2-18x+19|$ 10. 3
11. 14 km/hr 12. 120 m,40 m 13. 14 minutes 14. 25 15.(i) $x^2-6x+11=0$

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